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Q5

CO3 - AT1 - Analytical Problem Solving

20 marks

20 marks

- a) Symmetric
- b) Upper triangular
- c) Lower triangular
- d) Diagonal matrix
- e) Identity matrix
- f) Display all applicable classifications.

```

1 #include <stdio.h>
2 int main() {
3     int N,i,j,s=1,u=1,l=1,d=1,iden=1,x;
4     scanf("%d",&N);
5     int A[N][N];
6     for(i=0;i<N;i++) for(j=0;j<N;j++) scanf("%d",&A[i][j]);
7     for(i=0;i<N;i++) for(j=0;j<N;j++){
8         if(A[i][j]==A[j][i]) s=0;
9         if(i>j && A[i][j]!=0) u=0;
10        if(i<j && A[i][j]!=0) l=0;
11        if(i==1 && A[i][j]==0) d=0;

```

$$\begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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QUESTIONS

Q1

CO3 - AT1 - Analytical Problem Solving

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Q2

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Q3

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Q4

CO3 - AT1 - Analytical Problem Solving

20 marks

Q5

CO3 - AT1 - Analytical Problem Solving

20 marks

Q2 CO3 - AT1 - Analytical Problem Solving

20 marks

2. Given an array of integers, write a C program that:
i. Determines whether the array is sorted.
ii. If sorted, performs Binary Search.
iii. Otherwise, performs Linear Search.
iv. Displays the search method used and the number of comparisons performed.

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int n, target, i, comp = 0, found = -1, sorted = 1;
4     printf("Enter number of elements: \n");
5     scanf("%d", &n);
6     int arr[n];
7     printf("Enter %d integers: \n", n);
8     for (i = 0; i < n; i++) scanf("%d", &arr[i]);
9     printf("Enter element to search: \n");
10    scanf("%d", &target);
11    for (i = 0; i < n - 1; i++) {
12        if (arr[i] > arr[i + 1]) {
13            sorted = 0;
```

Input

5
10 20 30 40 50
40

Output

35°C

Mostly sunny

Search

ENG IN

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20 marks

Q5

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20 marks

Q3 CO3 - AT1 - Analytical Problem solving

20 marks

3. Store marks of N students in M subjects using a two-dimensional array. Generate a report containing:
a) Student topper
b) Subject topper
c) Class average
d) Subject-wise average
e) Students scoring above the overall average
f) Rank list without modifying the original data

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int N, M;
4     scanf("%d%d", &N, &M);
5     int a[100][100];
6     int total[100];
7     int i, j;
8     for (i = 0; i < N; i++) {
9         total[i] = 0;
10        for (j = 0; j < M; j++) {
11            scanf("%d", &a[i][j]);
```

Input

2 3
80 90 70
60 50 40

Output

35°C Mostly sunny

Search

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QUESTIONS

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Q4

CO3 - AT1 - Analytical Problem Solving

20 marks

Q5

CO3 - AT1 - Analytical Problem Solving

20 marks

Q4

CO3 - AT1 - Analytical Problem Solving

20 marks

4. Given N strings, write a C program to:
a) Sort them lexicographically.
b) Find the longest and shortest strings.
c) Count duplicate strings.
d) Display strings beginning with vowels.

Your Code

```
1 #include <stdio.h>
2 #define MAXN 100
3 #define MAXL 101
4 int mylen(const char *s){
5     int i=0;
6     while(s[i]) i++;
7     return i;
8 }
9 int mycmp(const char *a,const char *b){
10    int i=0;
11    while(a[i] && b[i] && a[i]==b[i]) i++;
12    if((unsigned char)a[i] < (unsigned char)b[i]) return -1;
13    if((unsigned char)a[i] > (unsigned char)b[i]) return 1;
```

Input

5
apple
orange
kiwi

Output

35°C

Mostly sunny

Search

ENG

IN

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